

The Effect of Private Pressure on Corporate Disclosures: Evidence from a Randomized Field Experiment

ABSTRACT

Section 1502 of the Dodd-Frank Act requires SEC filers to disclose their use of conflict minerals, with the goal of improving supply-chain transparency. We conduct a randomized field experiment to test whether private, outsider-initiated pressure transmitted through firms' grievance mechanisms affects mandatory conflict minerals disclosures. We anonymously alert firms to the presence of a U.S.-sanctioned entity in their supply chains. Treated firms are significantly more likely to modify their subsequent disclosures, either by removing the sanctioned entity from their reports or by providing an explanatory narrative disclosure. Firms receiving an illustrative disclosure example are more likely to explain rather than remove. Cross-sectional analyses show that disclosure responses vary with firms' external visibility and stakeholder exposure, consistent with reputational cost considerations. Many firms engage substantively with the inquiry. Overall, our findings provide causal evidence that private pressure, even absent public diffusion, shapes firms' disclosure choices in a setting characterized by managerial discretion and limited regulatory enforcement.

I. INTRODUCTION

Stakeholder pressure plays an important role in shaping corporate disclosure behavior. A large literature documents that external stakeholders, such as institutional investors, analysts, auditors, nongovernmental organizations (NGOs), and the media, can discipline managers and influence reporting choices by increasing scrutiny and reputational exposure (Lang and Lundholm 1993; Bushee, Matsumoto, and Miller 2003; Ball, Jayaraman, and Shivakumar 2012; Dyck, Volchkova, and Zingales 2008; Christensen et al. 2017; Rauter 2020). Much of this evidence focuses on public pressure, where information diffuses broadly and firms face heightened attention from capital markets, regulators, and other stakeholders.

Less is known about how firms respond to private pressure, such as concerns raised confidentially through corporate grievance mechanisms. Private inquiries are widespread in practice and are explicitly encouraged by regulators and international standard-setting bodies as part of corporate due-diligence frameworks, yet their effects are theoretically ambiguous. On the one hand, private pressure lacks public diffusion and may therefore be easier for firms to ignore,

particularly in settings characterized by weak enforcement. On the other hand, when private inquiries reference verifiable regulatory actions and are transmitted through institutionalized reporting channels, they may be perceived as credible and increase firms' expectations of future scrutiny, even in the absence of immediate public exposure. Whether such private pressure influences corporate disclosure remains an open empirical question.

We examine this question in the context of conflict minerals disclosure under Section 1502 of the Dodd–Frank Act. Section 1502 requires Securities Exchange Commission (SEC) registrants to conduct due diligence on the source of certain minerals (tin, tantalum, tungsten, and gold (3TG)) and to disclose the results annually in Form SD and, when applicable, a Conflict Minerals Report (CMR). The regulation was intended to improve supply-chain transparency and reduce the financing of armed groups in the Democratic Republic of the Congo (DRC) and surrounding regions. In practice, however, enforcement of the conflict minerals rule has been limited, and prior research documents substantial heterogeneity in firms' compliance and disclosure quality (Dalla Via and Perego 2018; GAO 2023).

At the same time, firms sourcing gold and other minerals remain subject to U.S. sanctions law administered by the Department of the Treasury's Office of Foreign Assets Control (OFAC). In March 2022, OFAC sanctioned African Gold Refinery (AGR), a gold refinery based in Uganda, for its alleged role in facilitating the illicit movement of gold from the DRC. According to the U.S. Treasury, over 90 percent of gold mined in the DRC is smuggled to neighboring countries, where it is refined and exported to global markets, generating substantial revenues for armed groups operating in eastern Congo (U.S. Treasury 2022). These groups—estimated to number more than 120—extract rents from artisanal mining and related activities, perpetuating conflict and

undermining regional stability (Sánchez de la Sierra 2020; Chang and Christensen 2023; Henn et al. 2024; Marchais et al. 2024).

The AGR sanction is particularly relevant for conflict minerals disclosures (CMD) for three reasons.¹ First, OFAC’s action directly aligns with the objectives of Section 1502, which seeks to increase transparency around mineral sourcing and discourage the financing of armed groups. Second, firms filing CMRs are required to conduct due diligence under nationally or internationally recognized frameworks, most commonly the Organisation for Economic Cooperation and Development (OECD) Due Diligence Guidance, which explicitly calls for firms to establish grievance mechanisms that allow stakeholders to raise concerns about sourcing risks.² Third, despite the sanction, many firms continued to disclose AGR as a smelter or refiner (SoR) in their CMRs, suggesting that disclosure choices in this setting involve managerial discretion and potential tradeoffs between transparency, reputational risk, and compliance costs.

We use this setting to examine whether private, outsider-initiated inquiries submitted through firms’ grievance mechanisms influence subsequent disclosure behavior. Ex ante, the direction of the effect is unclear. On the one hand, private inquiries may prompt disclosure changes by increasing the salience of potential regulatory or reputational risk. Prior research shows that firms respond to stakeholder scrutiny when reputational costs are salient (Shleifer and Vishny 1986; Lang and Lundholm 1993; Bushee et al. 2003; Dyreng, Hoopes, and Wilde 2016; Christensen et al. 2017; Rauter 2020), and that such costs can affect firm value and the cost of capital (Hombach,

¹ “Conflict minerals disclosures” (CMD) refers to the broader requirements for firms subject to Section 1502 of the Dodd-Frank Act, while the “conflict minerals report” (CMR) is the report that is only required as part of CMD if the firm determines that conflict minerals in its products may have originated from the DRC or adjoining countries. In this study, we use CMD and CMR interchangeably.

² The OECD’s guidelines are the most comprehensive international standard for what governments expect from multinational enterprises regarding human rights, disclosure, and due diligence. While not legally binding, they reflect consensus among over 50 countries and underpin the OECD’s mineral supply-chain due diligence framework.

Sellhorn, and Andreicovici 2018; Baik et al. 2024). A credible private inquiry may also alert firms that were unaware of the sanction or of AGR's continued presence in their disclosed supply chains.

On the other hand, firms may disregard private pressure if they perceive enforcement to be low or believe that disclosure changes are unnecessary.³ Some firms may operate under exemptions or licenses, while others may view a private inquiry as isolated and unlikely to generate follow-up scrutiny. Unlike public pressure, where media attention or activist campaigns amplify reputational consequences, private pressure lacks immediate diffusion and may therefore have limited bite. Whether firms alter their disclosures in response to such private inquiries is ultimately an empirical question.

Consistent with prior accounting research using randomized field experiments in real organizational settings (e.g., Sweeney and Roberts 1997; Free, Salterio, and Shearer 2009; Dennis and Johnstone 2018; Floyd and List 2016), we study this question using a randomized field experiment. We identify 234 firms that disclosed AGR as a SoR in their 2022 CMRs and randomly assign them to a control group or to one of two treatment conditions. Treated firms receive an anonymous inquiry submitted through their own grievance mechanisms that alerts them to OFAC's sanction of AGR and asks whether the firm has exemptions, whether there are undisclosed updates, and how the firm has addressed the sanctioned entity in its supply chain. One treatment condition includes only this inquiry, while the other additionally provides an illustrative disclosure example drawn from firms that voluntarily explained AGR's inclusion in earlier filings. This design allows us to examine whether private pressure alone affects disclosure behavior and whether providing a concrete disclosure model shapes how firms respond.

³ In April 2017, Acting SEC Chairman Michael S. Piwowar announced that a part of the conflict minerals due-diligence disclosures requirement would not be enforced following the U.S. Court of Appeals ruling (SEC 2017a, 2017b). No subsequent SEC priorities indicated a renewed enforcement (SEC 2023, 2024).

Treatment alerts were submitted through firms' existing grievance mechanisms, primarily via web-based ethics hotlines and, when such platforms were unavailable, via compliance- or ethics-related email addresses listed in firms' conflict minerals disclosures or codes of conduct. Consistent with prior research (Baloria, Marquardt, and Wiedman 2017), these mechanisms are designed to allow anonymous reporting by both internal and external stakeholders. Our submissions did not require identifying information, and we did not disclose our identity or institutional affiliation. In most cases, alerts were transmitted through third-party platforms that anonymize submissions before delivery to the firm, making it unlikely that recipients could infer the sender's identity. Section V and Appendix C provide additional implementation details.

The intervention is delivered after the end of the 2023 reporting year but before the May 31, 2024, filing deadline for 2023 CMRs. Historical filing patterns indicate that most firms finalize their conflict minerals disclosures in the weeks immediately preceding the deadline. Administering the inquiry approximately seven and a half weeks before the deadline provides firms with sufficient time to revise disclosures while limiting the scope for contemporaneous operational changes. At the same time, we acknowledge that some firms may be able to alter supply-chain relationships quickly. Importantly, however, CMRs are intended to reflect smelters and refiners used during the reporting year. As a result, even firms that sever ties shortly before filing may still be required to disclose AGR if it was used earlier in the year. Our analysis therefore focuses on observed disclosure responses and does not attempt to distinguish disclosure changes from rapid operational adjustments, which are unobservable to the public.

Because the inquiry is delivered privately and is not publicly disseminated, information diffusion is held constant by design. The intervention does not generate media coverage, investor attention, or regulatory action. This feature allows us to study disclosure responses in a setting

where private pressure operates primarily through perceived credibility and anticipated scrutiny rather than through public exposure. Importantly, while the inquiry does not increase realized dissemination, it may alter firms' expectations about the likelihood of future dissemination or follow-up by regulators or other stakeholders. As a result, firms may revise disclosures in anticipation of potential downstream scrutiny, even though the initial pressure itself remains private.

We examine firms' subsequent disclosures in their 2023 CMRs. We find that firms receiving a private inquiry are more likely to modify their disclosures to address the sanctioned entity, either by removing AGR from their SoR list or by retaining it while providing an explanatory note. Firms that receive an illustrative disclosure example are more likely to provide explanatory disclosure rather than removal. We also document substantial engagement with the inquiry: most treated firms respond, often indicating that they initiated internal reviews or supplier follow-ups. These findings suggest that firms take private inquiries transmitted through grievance mechanisms seriously, even in a regulatory environment characterized by limited enforcement.

We further examine cross-sectional variation in disclosure responses. Consistent with differences in anticipated reputational exposure, we find that firms facing greater public or ESG-related media scrutiny are more likely to respond by removing the sanctioned entity from their disclosures, while consumer-facing firms are more likely to respond by enhancing narrative explanation rather than removal. These patterns do not identify a specific causal mechanism but are consistent with firms tailoring disclosure responses to private pressure based on how different stakeholder audiences may evaluate those disclosures *ex post*.

Our study makes several contributions. First, we provide causal evidence that private, outsider-initiated pressure transmitted through firms' grievance mechanisms affects mandatory disclosure

behavior. This extends prior work that has largely focused on public pressure or insider whistleblowing by showing that credible private inquiries, delivered without public diffusion, can influence firms' disclosure choices. Second, we document that firms respond to private pressure through distinct disclosure adjustments, removing a sanctioned entity versus retaining it with narrative explanation, and that these responses vary with proxies for reputational exposure. These patterns are consistent with a reputational-cost framework in which credibility, rather than diffusion, drives disclosure responses when pressure is private. Third, we contribute to the growing literature using randomized field experiments in accounting by examining real disclosure outcomes in a regulatory setting characterized by managerial discretion and weak enforcement (Floyd and List 2016; Gassen and Muhn 2025).

Finally, by showing that grievance mechanisms can induce disclosure changes in the absence of strong SEC enforcement, we inform ongoing debates about the effectiveness of conflict minerals disclosure regulation (Christensen 2022). Consistent with prior evidence of weak enforcement, we document substantial noncompliance, including a declining number of firms filing CMRs and continued disclosure of a sanctioned entity in supply chains.⁴ Our findings are relevant to regulators and standard setters, including the SEC, OFAC, Government Accountability Office (GAO), Responsible Minerals Initiative (RMI), and OECD, by illustrating how disclosure mandates, grievance mechanisms, and private stakeholder scrutiny interact to shape firms' disclosure practices.

We acknowledge important limitations. Our analysis focuses on disclosure responses rather than verified changes in underlying supply-chain relationships. Although our empirical setting is narrow, we use the conflict minerals disclosure regime as a tractable laboratory to study how firms

⁴ The Government Accountability Office (GAO) reported a 24 percent decline in the number of firms filing CMDs, from 1,321 in 2014 to 1,005 in 2022, raising concerns about the effectiveness of the CMD requirement (GAO 2023).

adjust mandatory disclosures in response to private pressure when enforcement is limited and managerial discretion is substantial. Accordingly, our conclusions do not speak to the broader effectiveness of sanctions or disclosure mandates. Nevertheless, our findings suggest that grievance mechanisms represent an underexplored channel through which private stakeholders can influence corporate disclosure behavior.

II. BACKGROUND AND RELATED LITERATURE

Prior research shows that information pressure influences firms through information diffusion or credibility (Dyck, Volchkova, and Zingales 2008). Our setting isolates private pressure transmitted through firms' grievance mechanisms, holding diffusion constant by design and allowing us to focus on how credibility and anticipated scrutiny shape disclosure choices in a context with limited enforcement and substantial managerial discretion.

Conflict Minerals Disclosures and Sanctions

Section 1502 of the Dodd–Frank Act requires U.S. firms to conduct due diligence on the source of conflict minerals (i.e., tin, tantalum, tungsten, and gold (3TG)) and to disclose the results annually through Form SD and, when applicable, a Conflict Minerals Report (CMR). Following the OECD Due Diligence Guidance, firms must identify smelters or refiners (SoRs) in their supply chains and describe due diligence efforts. These disclosures combine standardized elements with narrative discussion and therefore involve considerable managerial judgment.

Enforcement of Section 1502 has historically been limited, and liability for CMR disclosures falls under Exchange Act Section 18. At the same time, firms remain subject to U.S. sanctions administered by the Office of Foreign Assets Control (OFAC). When OFAC sanctioned African Gold Refinery (AGR) on March 17, 2022, firms with exposure to AGR faced increased regulatory and reputational risk. Even absent immediate enforcement, continued disclosure of a sanctioned

entity may attract scrutiny from regulators (e.g., the SEC and OFAC), and other stakeholders, making disclosure choices a salient margin of response.

Private Pressure, Credibility, and Grievance Mechanisms

Information pressure can operate through diffusion and credibility (Dyck et al. 2008). Public pressure typically activates both channels, whereas private pressure transmitted through grievance mechanisms holds diffusion constant, allowing credibility to play a central role.

Grievance mechanisms—a core element of the OECD due diligence frameworks—function as early-warning systems for sourcing and compliance risks. Inquiries submitted through these channels are typically routed to legal, compliance, or ethics functions and evaluated alongside disclosure obligations. As a result, concerns raised through grievance mechanisms can carry institutional salience even when anonymous and not publicly disseminated.

This context distinguishes grievance-based private pressure from public activism and traditional whistleblowing. Unlike public pressure, private inquiries do not generate immediate media or investor attention. Unlike whistleblowing, they do not necessarily allege wrongdoing or trigger formal investigations. Instead, credible private inquiries, especially those referencing verifiable regulatory actions, may increase managers' expectations of ex post scrutiny, raising anticipated reputational or compliance costs associated with maintaining unchanged disclosures.

Related Literature

Our study relates to the literature on whistleblowing and internal reporting, which primarily examines how internal employees expose confirmed misconduct and how firms design and respond to reporting systems (e.g., Near and Miceli 1985; Treviño, Weaver, Gibson, and Toffler 1999; Baloria, Marquardt, and Wiedman 2017). Prior research shows that reporting effectiveness varies with system design and accessibility (Scheiber 2017; Soltes 2020), and that firms rely on a

range of internal and external stakeholders to surface compliance concerns. While reporting systems can uncover wrongdoing (e.g., Calderon-Cuadrado et al. 2009; Lee and Fargher 2013; Johansson and Carey 2016), existing evidence largely centers on internal allegations of realized misconduct. Our setting differs by examining anonymous, outsider-initiated inquiries submitted through grievance mechanisms that do not allege misconduct and occur prior to enforcement. Our design allows us to study preemptive disclosure responses to credible private pressure.

This paper also relates to the literature on media attention, stakeholder pressure, and disclosure behavior, which shows that public scrutiny disciplines managers through reputational concerns (e.g., Lang and Lundholm 1993; Bushee et al. 2003; Dyck et al. 2008; Christensen et al. 2017; Rauter 2020). Unlike this work, our setting holds diffusion constant, enabling us to isolate the role of credibility and anticipated scrutiny.

More broadly, our study relates to a long-standing tradition of field experimentation in accounting, particularly in management accounting, auditing, and organizational settings. Prior research has used field experiments conducted in real organizational environments to examine how accounting systems, controls, and professional norms influence behavior inside firms (e.g., Sweeney and Roberts 1997; Free, Salterio, and Shearer 2009; Kelly, Presslee, and Webb 2017; Dennis and Johnstone 2018; Cronin, Erkens, Schloetzer, and Tinsley 2021). This literature shows the value of randomized or quasi-randomized interventions conducted in natural institutional settings for identifying causal effects (List, Sadoff, and Wagner 2011). Building on this methodological tradition, more recent studies apply randomized field experiments to examine firms' responses to regulatory nudges and information interventions (e.g., Floyd and List 2016; Gassen and Muhn 2025), as well as disclosure responses to external pressure such as NGO or media-driven scrutiny (Belnap 2023) and retail investor activism (Wong, Yu, Zhang, and Zhang

2024). These studies primarily examine settings in which pressure operates through public visibility and information diffusion.

By contrast, existing research provides limited evidence on how private, outsider-initiated pressure transmitted through institutionalized grievance mechanisms affects disclosure behavior in mandatory reporting settings. Our study builds on this distinction by examining whether private inquiries influence disclosure choices through credibility and anticipated reputational costs, holding public diffusion constant by design.

III. A REPUTATIONAL-COST FRAMEWORK FOR PRIVATE PRESSURE

To interpret how private pressure affects disclosure behavior, we develop a simple reputational-cost framework. Unlike public pressure that operates through widespread information diffusion, the inquiry in our setting is delivered privately through firms' grievance mechanisms and is not publicly disseminated. Diffusion is therefore held constant by design. Following Dyck, Volchkova, and Zingales (2008), we model managerial responses as reflecting expectations about reputational and enforcement costs. In our setting, reputational costs are firm-level and arise from anticipated scrutiny by relevant audiences involved in compliance and disclosure oversight, including regulators, legal counsel, investors, and other stakeholders, rather than from managers' personal labor-market concerns. Private inquiries therefore matter to the extent that they are perceived as credible by these audiences. The framework below formalizes this logic and motivates our empirical hypotheses.

Reputational Incentives and Disclosure Decisions

Consider a manager deciding whether to revise an existing disclosure when new information becomes available. Revising a disclosure by removing a supplier, providing an explanatory note, or conducting additional due diligence may impose private costs, including managerial time, legal

review, and internal investigation costs. Maintaining the status quo, however, may expose the firm to reputational or regulatory risk if the disclosure is later perceived as misleading or noncompliant. Following Becker (1968) and Dyck, Volchkova, and Zingales (2008), a disclosure revision occurs if the expected private cost of revision is outweighed by the expected cost of inaction:

$$\begin{aligned} E(\text{Cost of revision}) &< E(\text{Reputational cost}) + E(\text{Enforcement cost}) \\ &= E(\text{Cost of revision}) < \sum_i p_i RC_i + \pi P, \end{aligned} \tag{1}$$

where RC_i is the reputational cost with respect to audience i , and p_i is the probability that audience i becomes aware of, believes, and negatively evaluates the firm's disclosure decision. Relevant audiences include internal and external actors such as regulators (e.g., the SEC and OFAC), legal counsel, compliance officers, auditors, and other stakeholders, including investors, who may evaluate the firm's disclosure decisions. Expected enforcement cost is given by πP , where π is the perceived probability of regulatory scrutiny and P is the associated penalty. In our setting, this tradeoff arises when firms determine how to disclose supply-chain exposure to a sanctioned entity in their CMRs.

Diffusion versus Credibility in Private Pressure Settings

Information pressure can operate through two distinct channels: diffusion and credibility (Dyck et al. 2008). Diffusion affects how widely information spreads, while credibility affects whether information is believed and therefore imposes reputational costs. Because the inquiry in our experiment is delivered privately, diffusion is largely held constant by design. The inquiry does not generate immediate public awareness, media coverage, or investor attention. Instead, its effect depends primarily on credibility, that is, whether the inquiry is sufficiently legitimate to warrant concern about future scrutiny by relevant audiences. Credibility affects p_i not by expanding the audience, but by increasing the probability that a small but relevant set of actors such as regulators

(e.g., SEC and OFAC), legal counsel, or compliance officers will treat the issue seriously if reviewed ex post.

How Private Inquiries affect Reputational and Enforcement Costs

A private inquiry can affect the expected cost of inaction through three related channels. First, it can increase perceived awareness among relevant audiences by alerting internal compliance or disclosure gatekeepers, raising the likelihood that the disclosure will be scrutinized in the future. Second, it can increase the perceived credibility of the concern when it references verifiable regulatory actions and is transmitted through an institutionalized reporting channel, thereby increasing RC_i . Third, it can increase perceived enforcement risk by raising the perceived probability π that regulators or other stakeholders may question the disclosure. These effects operate through expectations rather than realized enforcement.

Implications for Disclosure Behavior

This framework yields two implications. First, private pressure can induce disclosure changes even when firms lack sufficient time to alter real supply-chain activities. When the cost of revising a disclosure is low relative to the expected reputational cost of inaction, firms may respond by modifying disclosure content rather than making immediate operational changes. Second, heterogeneity in responses should reflect differences in firms' sensitivity to credibility and reputational exposure, such that firms with more formalized compliance structures or greater legal oversight respond more strongly to private inquiries.

Proposition

Holding public diffusion constant, a firm is more likely to change its subsequent disclosure to address a potential sanction or compliance concern when a private inquiry increases the perceived

credibility and salience of that concern, thereby raising the expected reputational and compliance costs of inaction.

Scope and Limitations

This framework is not intended to identify a specific psychological or organizational process through which managers respond to private alerts. Rather, it provides a parsimonious economic structure linking private stakeholder pressure to disclosure behavior through expectations about reputational and enforcement costs. The empirical analysis therefore focuses on whether private pressure affects disclosure choices and provides suggestive evidence consistent with a credibility-based mechanism, without claiming formal mediation.

IV. HYPOTHESIS DEVELOPMENT

The framework developed in Section III predicts that, holding public diffusion constant, private inquiries submitted through firms' grievance mechanisms can increase the expected reputational and compliance costs of maintaining an unchanged disclosure. In our setting, this tradeoff arises in firms' CMRs, which disclose supply-chain exposure to smelters or refiners subject to regulatory scrutiny.

On March 17, 2022, OFAC sanctioned AGR. Firms that continued to disclose AGR as a refinery in their CMRs therefore faced potential reputational and compliance risk associated with sanction violations or the perception of inadequate due diligence. Our inquiry, delivered privately through firms' grievance mechanisms after the reporting year-end but before the CMR filing deadline, represents a form of private stakeholder pressure directed at this disclosure choice.

Because the inquiry was sent after the reporting year-end, firms had limited scope to alter underlying supply-chain relationships within the remaining time before filing. However, firms retained discretion over how to disclose supply-chain exposure in their subsequent CMRs. As a

result, any response to the inquiry is most likely to occur through disclosure adjustments rather than immediate operational changes. Some firms may choose not to respond, for example if they believe enforcement risk is low or if they operate under an exemption or license. Others may respond by modifying their disclosures to mitigate expected reputational or compliance costs.

Consistent with the reputational-cost framework developed in Section III, a private inquiry that is perceived as credible may raise the expected cost of inaction by increasing the likelihood of future scrutiny by relevant audiences, such as regulators, legal counsel, or investors. When the expected cost of revising a disclosure is low relative to the expected reputational and enforcement costs of maintaining the status quo, firms may respond by modifying how the sanctioned entity is disclosed in the CMR. Such modifications may include removing the sanctioned entity from the reported SoR list or providing explanatory disclosure that contextualizes the firm's exposure or due-diligence efforts.

Importantly, these disclosure changes reflect firms' reporting responses to private pressure and do not necessarily imply contemporaneous changes in underlying supply-chain relationships. Rather, they capture firms' efforts to manage expected reputational and compliance risks associated with maintaining an unchanged disclosure. Accordingly, we test whether private pressure applied through grievance mechanisms affects subsequent disclosure behavior.

***H1:** Firms that receive a private inquiry through their grievance mechanism are more likely to modify their subsequent conflict minerals disclosures to address the sanctioned entity.*

V. THE EFFECTS OF PRIVATE PRESSURE

Sample of Firms with African Gold Refinery

Recall that firms must disclose their use of conflict minerals by submitting a CMR along with Form SD as outlined in Section II (SEC 2012).⁵ To identify relevant disclosures, we use Intelligize to search 2022 SEC filings for references to “African Gold Refinery” or “AGR.”⁶ We focus on 2022 because it is the first reporting period following OFAC’s sanction of AGR on March 17, 2022, giving firms sufficient time to account for the sanctions before the May 31, 2023 filing deadline.

We compile all Form SDs and CMRs, validating AGR’s inclusion through searches within each CMR. As CMRs typically list SoRs identified via firms’ reasonable country of origin inquiries and the RMI’s conflict minerals reporting template, we confirm AGR’s presence in each firm’s SoR list.⁷ We verify the presence of AGR on the firm’s 2022 SoR list and identify seven firms that provided rationales for AGR’s inclusion on their 2022 SoR despite the sanctions. We present details on these seven firms in Appendix B. We exclude these seven firms with explanations and three pilot firms, leaving a final sample of 234 firms (Table 1, Panel A). Using the Fama-French 12 industry classification, our sample is concentrated in business equipment (31 percent) and manufacturing (30 percent), followed by healthcare (13 percent) and consumer durables (11 percent) (Table 1, Panel B). We supplement these data with revenues, assets, firm age, market value, and sales growth from Compustat and CRSP; regulatory violations from the Violations Tracker database; coverage of supplier social impact measures from Asset 4; coverage of employee satisfaction by Glassdoor; the market reaction at the time of the OFAC sanctions as a proxy for prior awareness (Chen, Goldstein, and Jiang 2007); and public attention (i.e., EDGAR

⁵ Item 1.01 describes the CMR and directs readers to the CMR in the exhibits section of Form SD.

⁶ Intelligize compiles SEC disclosure documents, including filings, comment letters, and SEC interpretations, into a platform that is easy to use. See <https://www.intelligize.com/about/>

⁷ The RMI promotes responsible sourcing by providing third-party audits of SoRs, verifying compliance with global standards, and offering a CMR template (RMI 2024). See <https://www.responsiblemineralsinitiative.org/about/rmi-initiative/>

downloads of recent specialized disclosures during the year prior to our intervention). We also hand-collect the number of gold smelters listed in each 2022 CMR and use the inverse count as a proxy for the proportion of AGR association.⁸ Overall, our sample is intentionally limited to firms that disclosed AGR as a smelter or refiner. It identifies a setting in which private stakeholder pressure through grievance mechanisms is feasible. We interpret our findings within this institutional scope.

Data Collection

Next, we collect data on grievance mechanisms, a core component of the OECD's risk-based due diligence framework, which requires firms to establish company- or industry-level mechanisms as early-warning systems (OECD 2016). For each firm, we examine the CMR to determine whether and how a grievance mechanism is referenced. Some firms provide specific hotline details (weblink, phone number, or email) in the CMR, whereas others only acknowledge its existence. When such information is absent, we review the firm's online code of conduct to retrieve it, if publicly available. We also collect the name, title, and phone number of the officer responsible for signing each firm's Form SD. This approach allows us to evaluate both the accessibility and transparency of grievance mechanisms as part of firms' conflict minerals compliance efforts.

Designing the Field Experiment

We design our study to answer the following questions: (1) How do firms respond to private pressure exerted via their grievance mechanisms? (2) Do firms change their subsequent CMDs following a private inquiry through a grievance mechanism about a sanctioned entity in their supply chain? If so, how? To conduct our inquiry, we design two instruments. We design the

⁸ Two firms did not disclose gold smelters in a countable format. To preserve the sample, we impute the median number of gold smelters (175) for these firms.

instruments to probe firms about their responses to OFAC's March 17, 2022, sanctions on AGR. Figure 1 presents a timeline of events related to the AGR sanctions and our intervention. The instrument for our inquiry informs the treated firms of AGR's presence in their supply chains and provides an explanation of the OFAC sanctions and the reasons behind them, supported by a URL link to the OFAC press release. Then we pose three questions: (1) Does your firm have any exemptions (e.g., from OFAC or the Department of Justice) that allow continued business dealings with AGR? (2) Are there any updates to your company's disclosures regarding AGR that we might be unaware of? (3) If no updates are available, could you explain your company's actions in response to this situation?

The first question helps us identify whether firms received exemptions that might influence their disclosures. OFAC regulations prohibit U.S. persons or entities from transacting with sanctioned parties *unless* authorized by a general or specific license (U.S. Treasury 2022). Firms with such exemptions or licenses might present CMDs differently from those without. The second question explores whether the reporting entity updated its CMR after the initial 2022 filing regarding AGR but did not make them publicly available. The third question is open-ended, allowing firms to disclose their ongoing efforts to address the involvement of a sanctioned entity in its supply chain. Instrument 1 concludes with these three questions and is detailed in Appendix C. Instrument 2, also in Appendix C, adds an illustrative disclosure: Tesla's acknowledgment of the AGR sanction in its 2022 CMR. In constructing the second treatment instrument, we draw on disclosure language used by the seven firms that explicitly provide rationales for the continued inclusion of AGR on their smelter or refiner lists in their 2022 conflict minerals reports, despite the OFAC sanction. The seven firms include Tesla, Ford Motor Co., Amphenol Corp., Ormat Technologies Inc., Canon Inc., Aptiv PLC, and Allot Ltd. We reference Tesla as one illustrative

example because its disclosure language is representative of this broader set and provides a clear example of the rationale used by firms. Apart from its prominence in the business press, Tesla does not differ systematically from the other firms that provided similar explanations and plays no special role in the empirical analysis. Instrument 2 enables us to test whether information provision of another entity's disclosure could influence other firms' intermediate responses to our inquiry or their subsequent conflict minerals disclosures.⁹

To examine the effect of private pressure on conflict minerals disclosures through grievance mechanisms, we randomly assign 234 firms into three experimental arms: a control group (78 firms) and two treatment groups, with 78 firms receiving Instrument 1 and 78 firms receiving Instrument 2. The allocation of two-thirds of firms to the treatment arms was chosen to improve statistical precision and to facilitate comparisons across the two treatment instruments. This allocation does not address nonrandom attrition. We acknowledge that the sample size is modest. Given the number of treated and control firms, the design is powered to detect economically meaningful disclosure responses but may lack power to detect very small effects. Accordingly, null results, particularly in comparisons between the two treatment instruments, should be interpreted with caution. This limitation reflects the size of the underlying population of firms disclosing African Gold Refinery and cannot be addressed through additional data collection.¹⁰

Treatment Delivery, Reporting Channels, and Anonymity

Treatment alerts were submitted using firms' pre-existing grievance mechanisms, consistent with how firms invite compliance concerns from both internal and external stakeholders. For

⁹ Haaland, Roth, and Wohlfart (2023) discusses applications of information provision experiments. Our treatment varies one feature of the information set provided to the two groups.

¹⁰ A simple two-sample power calculation indicates that, with 156 treated firms and 78 control firms, the design achieves approximately 80 percent power to detect a standardized effect size of about 0.39 at the five percent significance level.

129 of the 156 treated firms (82.7 percent), we submitted alerts through web-based ethics hotlines provided in firms' conflict minerals reports or codes of conduct. These hotlines are typically operated by third-party providers and allow anonymous submission without requiring identifying information. For the remaining 27 firms (17.3 percent), which did not maintain functional web-based hotlines, we submitted alerts via compliance, ethics, or hotline-designated email addresses disclosed by the firm.

Among the web-based submissions, the vast majority were transmitted through third-party incident-management platforms (e.g., EthicsPoint, Convercent, Mitrastech Syntrio), which anonymize reports before forwarding them to the firm. These platforms explicitly advertise anonymity protections consistent with firms' stated whistleblowing and grievance policies. Prior research documents that such mechanisms are designed to facilitate anonymous reporting by both employees and external stakeholders (Baloria, Marquardt, and Wiedman 2017). Our submissions did not require login credentials, IP addresses, or personal identifiers, and we did not disclose our identity or institutional affiliation in any communication.

Email submissions were used only when no web-based reporting system was available. These messages were sent from a newly created, anonymous email (Gmail) account that did not contain identifying information. Emails were directed to addresses explicitly designated for ethics, compliance, whistleblowing, or related concerns, following firms' own instructions for submitting reports. In these cases, the reporting channel mirrors how external stakeholders would ordinarily contact the firm regarding compliance concerns. Our choice to remain anonymous is driven by our intent to observe how our intervention impacts firm disclosures, regardless of the source of information about AGR. Firms' responses might differ if they know the source of the inquiries (Haaland et al. 2023). Our inquiry into the presence of AGR on treatment firms' 2022 SoR lists

and OFAC sanctions is factual, ensuring that our approach is transparent and truthful. Maintaining anonymity helps elicit unbiased responses from the treated firms; however, it limits us to inferring their perceptions of the inquirer's identity based on their intermediate responses.

Firm responses are consistent with the alerts being perceived as anonymous and credible. In multiple cases, firm representatives explicitly acknowledged the reporter's anonymity and indicated that they would proceed with internal review or investigation based on the information provided. A firm requested additional details while offering mechanisms to continue communicating anonymously through the reporting platform. Only one firm encouraged disclosure of identifying information, which we declined. Overall, we find no evidence that firms inferred that the alerts originated from academic researchers rather than from external stakeholders.

Appendix C provides detailed documentation of the reporting channels used, including the distribution of alerts across third-party platforms, proprietary web portals, and email-based submissions.

Timing of the Intervention

We alert the treatment firms seven and a half weeks before the CMD filing deadline. If treated firms modify their disclosures, by removing AGR from their SoR lists or referencing the AGR sanctions, relative to control firms, this suggests that our intervention increased the salience of AGR. The timing of the intervention was chosen to occur after the end of the 2023 calendar year but before the deadline for filing CMRs (May 31, 2024). This design choice was intended to ensure that any observed response would occur through firms' subsequent disclosure decisions rather than contemporaneous changes in underlying supply-chain relationships. At the same time, we sought to allow firms sufficient time to revise disclosures prior to filing their CMRs.

To determine a reasonable window to provide firms with adequate opportunity to update disclosures, we examined historical filing behavior. In 2022, 92 percent of firms filed within the two weeks preceding the May 31, 2023 deadline, and 81.7 percent filed within the final seven days. Similarly, in 2023, 92.7 percent filed within the final two weeks, and 80.7 percent filed within the final seven days. Filing patterns in 2021 are comparable, with 83.6 percent of firms filing within seven days of the deadline. Over these three years, fewer than two percent of firms filed more than one month before the deadline. These patterns indicate that firms typically finalize CMRs close to the deadline, and so we determined that a period of about eight weeks would be long enough. Ultimately, we submitted our inquiries on April 9, 2024, 52 days before the May 31, 2024, filing deadlines for 2023 CMDs.

This timing choice has both advantages and limitations. An advantage is that it allows us to observe disclosure responses while limiting the scope for contemporaneous operational adjustments. A limitation is that some firms may be able to alter supply-chain relationships quickly after receiving the inquiry. However, the CMR lists smelters and refiners in the firm's supply chain used during the reporting year (2023). As a result, even if a firm severed ties with the sanctioned entity (AGR) shortly before filing, that entity may still appear in the CMR if it was used earlier in the reporting period. Accordingly, our analysis focuses on observed disclosure changes rather than attempting to distinguish disclosure responses from rapid operational adjustments.

Responses to Private Inquiries

We first conduct a small pilot test with three firms using their third-party online reporting systems, submitting inquiries with Instrument 1 (Appendix C). Consistent with prior field experiments in accounting and finance, the purpose of the pilot is not hypothesis testing but procedural validation: specifically, to verify message deliverability, confirm that inquiries are

successfully routed through grievance mechanisms, and assess whether firms acknowledge or process anonymous submissions as intended (e.g., Floyd and List 2016; Soltes 2020; Gassen and Muhn 2025). We deliberately limit the pilot to three firms to minimize learning, contamination, or spillover effects while ensuring that the intervention is operationally feasible before full deployment. The pilot provides preliminary insights into feasibility. We then submit inquiries to the remaining treatment firms through their online reporting systems or, when unavailable, via designated reporting email addresses.¹¹ For each submission, we record the reporting channel, submission and response dates, and whether firms acknowledged receipt or provided follow-up communication. We classify responses on a five-point scale: 0 = no response; 1 = automated system reply; 2 = human response with no commitment to action; 3 = human response indicating an investigation; and 4 = human response offering substantive answers.

CMR Disclosures After AGR Inquiry

We begin our empirical analysis by examining the 2023 CMRs filed (May 31, 2024, deadline) after sending our inquiries to the treatment firms. Our goal is to determine whether firms that receive inquiries through a private channel change their disclosures. These tests rely on comparative analyses of CMR disclosures from both treatment and control firms. We estimate the following linear probability model:

$$Disclosure_i = \alpha_0 + \beta_1 Treat_i + \sum \beta Controls_i + \varepsilon_i \quad (2)$$

Disclosure is either (i) *Removed_AGR*, a binary variable equal to one if a firm's 2023 CMR removes AGR either from its SoR list (or elsewhere in the report) and zero if AGR is still mentioned, or (ii) *Explained_AGR*, a binary variable equal to one if a firm's 2023 CMR includes

¹¹ Some firms use third-party platforms, including NAVEX and Convercent, whereas others use their own platforms.

AGR on its SoR list and provides an explanation or reference to the AGR sanctions in the firm's CMR (in the main text or footnotes) and zero otherwise. *Treat* equals 1 (0) if the firm is in the treatment (control) group. We differentiate the information provision to see whether providing an example of an explanation for AGR in the CMR increases a firm's perception of its value and consequently causes the firm to also explain why AGR is in its own CMR. We include the following control variables: *Legal Signor* and *CFO Signor*, indicators equal to one if the Form SD is signed by legal counsel or the CFO, respectively; *Revenue*, the firm's 2022 total revenue; *Market Value*, the natural logarithm of firm's 2022 market capitalization; *Sales Growth*, the percentage change in sales from 2021 to 2022; *Firm Age*, the number of years since listing on CRSP as of December 2022 that the firm has been on CRSP; *Proportion*, the fraction of gold smelters constituted by African Gold Refinery in 2022; *Asset 4 Coverage*, a binary variable equal to one if the firm is covered by Asset 4 on supplier social impact; *Violation*, a binary variable equal to one if the firm has a violation in the last three years, as documented by the Violation Tracker database from Good Jobs First, and zero otherwise; *Glassdoor*, a binary variable equal to one if the firm is covered by Glassdoor employee satisfaction ratings, and zero otherwise; and *Prior Awareness*, the firm's stock market reaction at the time of the U.S. Department of Treasury's sanctioning of African Gold Refinery, consistent with managers learning from stock prices as in Chen, Goldstein, and Jiang (2007).

Descriptive Statistics and Results of the Randomized Field Experiment

Table 2 presents the descriptive statistics for our key variables. Panel A indicates that firms in our sample are well-established, with a mean (median) firm age of 30 (27) years. These firms are generally midsized, with an average (median) market value of \$22 (\$2.9) billion. On average, firms

have at least one violation in the Violation Tracker database during the three years prior to our sample period.

Panel B presents the descriptive statistics for firms' grievance mechanisms and SD signors. The results indicate a significant use of grievance mechanisms with 62 percent of the firms including grievance mechanisms in their CMRs. In addition, 83 percent of the firms include hotline information in their codes of conduct, and 81 percent provide website-based hotlines suggesting a high level of accessibility to grievance systems. Over 46 percent of the sample CMRs are signed by legal counsel, which could imply a higher degree of formality and legal oversight in the grievance process, 35 percent are signed by CFOs, and the rest by CEOs and other executives, indicating widespread management involvement in CMDs. We also examine the ease of accessibility of firms' grievance mechanisms using a determinants test. Whether a firm includes its grievance mechanism in its CMR could signal the accessibility of the reporting system. We find that CMRs with Legal or CFO Signors are less likely to contain grievance mechanisms highlighting the role of the signor.

Table 2 Panel C shows the descriptive statistics of the treatment and responses. Two-thirds of our sample received either Treatment 1 or 2. By June 3, 2024, 125 firms (80 percent) had responded to our inquiry, with an average response time of 15 days and an average length of 106 words. We present examples of firm responses in Appendix D. Notably, in an untabulated analysis, only 2.9 percent of firms indicated that they discouraged anonymous reporting, suggesting a general openness to receiving anonymous feedback. This result highlights the responsiveness and engagement level of the firms with a grievance mechanism in place. Table 2 Panel D shows the descriptive statistics of the treatment and disclosure consequences. 12 firms (eight in treated and

four in control) did not file their Form SD by June 3, 2024, leaving a final sample of 222 firms.¹² Among filers, 23.4 percent removed AGR from their Form SDs, while 15.3 percent explained why the sanctioned entity was included on their SoR list.

Table 3 presents summary statistics on intermediate firm responses. Panel A provides the types of responses: 34 percent of the treated sample offered a human response indicating the initiation of an investigation, 31 percent provided specific answers to some of our questions, 20 percent did not respond, 10 percent returned an automatic response shortly after we submitted our inquiry, and the remaining 5 percent provided a response that neither addressed our questions nor committed to investigate AGR in their supply chains. This distribution suggests varying levels of engagement with our inquiry, with a significant portion investigating the issues raised but some not responding at all.¹³ None of the intermediate responses suggest other stakeholder concerns regarding the reporting of AGR. Several firms that responded with an investigation indicated that they would urge suppliers to discontinue sourcing from AGR and, if necessary, suspend relationships with the suppliers. For example, one firm responded, “... *of the two suppliers that reported a connection to AGR, one has now ended that relationship in part due to our urging and the other has confirmed that neither of the two products we currently purchase from them contain any gold at all, let alone any gold purchased from AGR.*” Overall, these responses reflect the firms’ commitment to regulatory compliance, ethical sourcing, and transparency in supply chains. Panel B reports the timeliness and length of the responses. The results indicate that automatic responses are typically

¹² Specific reasons for not filing were acquisitions (six firms), delisting (one firm), and financial distress (one firm). We are uncertain why the other four firms did not file because of the absence of public information on the matter.

¹³ 3.2 percent of the treated firms faced obstacles to responding, including bankruptcy and acquisition (one firm), internal-only reporting systems, and invalid email addresses (four firms). Alternative outreach to three of the firms yielded no responses.

brief (mean 45 words) and nearly immediate (0.33 days), whereas human responses are more detailed (53–184 words) and slower, with an average response time of 15–20 days.

Impact of Intervention on Subsequent Disclosures

Table 4 presents the results of our intervention’s impact on subsequent CMDs. In column (2), treatment is positively associated with the likelihood that a firm removes AGR from its 2023 SoR list, although the effect is marginally significant at the 10 percent level ($\beta = 0.107$, t -statistic = 1.845) and only when controls are included. This implies that notified firms were 10.7 percentage points more likely to remove AGR from their disclosures than control firms. Columns (3) and (4) examine the firm’s likelihood of explaining the presence of AGR in its 2023 CMR. The results show a significant positive effect: treated firms were significantly more likely than controls to provide additional disclosures about AGR, with column (4) showing a 21.8 percentage point increase ($\beta = 0.218$, t -statistic = 5.116). Figure 2 presents the number of treatment and control firms that filed specialized disclosures with AGR over time. It illustrates the plausibility of the parallel trends assumption and that AGR disclosures declined more steeply among treated firms than among control firms in 2024. This finding suggests that private pressure can influence firms’ disclosure choices.¹⁴ Taken together with results in columns (1) and (2), our findings imply that treated firms adjust their CMDs based on the information we privately provide to them.

Several intermediate responses highlight firms’ reliance on suppliers and industry surveys to compile SoR lists. For example, one firm explained that its supply chain involves multiple tiers between the company and its smelters, requiring dependence on suppliers to provide information

¹⁴ In untabulated analyses, we conduct a randomization test using the *ritest* package (Heß 2017) on STATA to check the robustness of our main results in Table 4 Columns (1) and (3). By permuting the treatment assignments 10,000 times, we obtained p-values of 0.154 and <0.001 respectively. The p-values suggest that the observed treatment effect is highly unlikely to be due to chance, thereby supporting the validity of our randomized field experiment’s findings.

on the origins of tin, tantalum, tungsten, and gold (3TG). Another firm shared its interim CMR template, noting that it was still collecting compliance data and that its final 2023 CMR was not yet available.

Treatment Heterogeneity: Disclosure Removal versus Explanation

We further examine the impact of treatment on subsequent CMDs by distinguishing between the treatment types. Table 5 presents the results of regressions that include both *Treat 1* and *Treat 2* as key variables. Column (2) shows a marginally significant coefficient for *Treat 1* ($\beta = 0.121$, t -statistic = 1.722), indicating that firms that receive inquiries without a disclosure example are somewhat more likely to remove AGR from their SoR lists relative to controls. Although *Treat 2* also yields a positive coefficient, it is not statistically significant at conventional levels. These findings suggest that providing an illustrative disclosure example did not increase firms' propensity to remove AGR. One possible explanation is that the disclosure example may have inadvertently signaled that retaining AGR could still be acceptable if accompanied by sufficient narrative explanation. Thus, firms in *Treat 2* may have opted to retain AGR while enhancing narrative justification in their CMRs rather than removing it.

Columns (3) and (4) present the results of the likelihood that firms explain the presence of AGR in their 2023 CMR. Both treatments are associated with a significantly higher probability of firms providing such explanations, in models both with and without control variables.¹⁵ The coefficient estimates and t -statistics for *Treat 2* are larger than those for *Treat 1*, suggesting that the disclosure example provided in *Treat 2* likely served as a useful model that other firms chose

¹⁵ In total, 34 firms in our sample explain the sanction on AGR. Among these, two are in the control group, ten in the *Treat 1* group, and 22 in the *Treat 2* group. Appendix E provides examples of seven firms in the *Treat 2* group that provide rationales in their CMR for AGR's inclusion in their 2023 list of smelters and refiners, despite the sanctions, after we provided information about the sanctioned entity.

to emulate. Consistent with this interpretation, ten *Treat 2* firms incorporated language identical to that in our inquiry when explaining AGR in their 2023 CMRs.

One explanation for the stronger treatment effects we observe for explanatory disclosures is that adding narrative context is less costly than removing a sanctioned entity from the smelter or refiner list, given the structure of conflict minerals reporting. This interpretation is consistent with the disclosure-versus-recognition literature, which shows that information disclosed in notes is less salient and incorporated less immediately into users' judgments than comparable recognized items (Clor-Proell and Maines 2014; Michels 2017). In our setting, private pressure transmitted through grievance mechanisms therefore appears to shift firms toward lower-cost narrative adjustments when full removal is more burdensome.

Cross-Sectional Variation in Disclosure Responses

We next examine whether firms' disclosure responses to private pressure vary systematically with external visibility, stakeholder scrutiny, and consumer exposure. These analyses are motivated by the reputational-cost framework developed in Section III, which predicts stronger disclosure responses when firms face higher expected reputational or compliance costs.

Tables 6 and 7 examine heterogeneity based on public and media attention. Table 6 shows that disclosure responses vary with public attention, measured by above-median downloads of firms' conflict minerals reports in the pre-period. Column (2) shows that the interaction between treatment and public attention ($Treat \times Attention$) is positive and marginally significant for removal outcomes ($\beta = 0.222$, $t = 1.829$), while the main effect of treatment is small and insignificant. Economically, treated firms with higher public attention are approximately 22 percentage points more likely to remove AGR from their subsequent smelter or refiner (SoR) lists relative to

similarly visible control firms. In contrast, column (4) shows no significant interaction effect for explanatory disclosures.

Table 7 presents complementary evidence using exposure to negative ESG-related media scrutiny, measured using RepRisk. In column (2), the interaction between treatment and RepRisk-based scrutiny ($Treat \times Media\ Attention$) is positive and statistically significant for removal outcomes ($\beta = 0.238, t = 2.103$), implying that treated firms facing elevated ESG-related scrutiny are approximately 24 percentage points more likely to remove AGR relative to control firms with similar scrutiny. As in Table 6, column (4) shows no corresponding interaction effect for explanatory disclosures. Together, these results indicate that private pressure is more effective in inducing disclosure removal when firms face heightened public or media scrutiny and therefore anticipate greater reputational costs from continued association with a sanctioned entity.

Table 8 examines heterogeneity in disclosure responses by business-to-consumer (B2C) status. The interaction between treatment and B2C status ($Treat \times B2C$) is negative and insignificant for removal outcomes but positive and statistically significant for explanatory disclosures ($\beta = 0.491, t = 2.582$). Although B2C firms exhibit a lower baseline propensity to provide explanations absent treatment, the positive interaction indicates that consumer-facing firms respond more strongly to private pressure by enhancing narrative disclosure. Economically, treated B2C firms are approximately 49 percentage points more likely to explain the continued presence of AGR relative to treated non-B2C firms, consistent with reputational concerns that emphasize justification and transparency in consumer-facing settings.

Taken together, the cross-sectional results in Tables 6–8 indicate that firms' disclosure responses to private pressure vary systematically with factors related to reputational exposure. Firms facing greater public or media scrutiny are more likely to respond by removing a sanctioned

entity from their disclosures, whereas consumer-facing firms are more likely to retain the entity while providing additional explanation. These patterns are consistent with the reputational-cost framework, which predicts that firms adjust disclosure choices in response to private pressure based on how potential audiences may evaluate those disclosures *ex post*.

In untabulated analyses, we examine whether disclosure responses to private inquiries vary with firms' reporting-channel structure, focusing on whether a firm's grievance mechanism includes an email option. We find that the removal response to private pressure is attenuated for firms whose grievance mechanisms include email as one of the available channels. This result is consistent with differences in the perceived salience or credibility of private inquiries transmitted through less formal communication channels, and it further supports the interpretation that the effectiveness of private pressure depends on how the inquiry is received rather than on public diffusion.

Potential Spillovers and Robustness

We acknowledge the potential for information spillovers from the treatment group to the control group. For example, after treating General Electric Co., we observed a similar removal of AGR in the subsequent disclosure of its spin-off, GE Healthcare Technologies Inc. Additionally, two firms in the control group included explanations related to the sanctions on AGR, further suggesting the possibility of spillover. Such effects may occur if employees responsible for CMRs in treated firms share information about our inquiry with their counterparts in control firms, given the close-knit nature of the responsible sourcing community. These interactions could have led to the unintended dissemination of information, potentially influencing the behavior of firms in the control group. These spillovers could violate the Stable Unit Treatment Value Assumption (SUTVA) (Angrist and Pischke 2009). However, any such spillovers, if present, would strengthen

the credibility of our findings. If the private alerts heightened compliance salience for control group firms to behave more like treated firms, the resulting reduction in observable differences between the two groups would bias our estimates toward zero. This conservative bias decreases the likelihood of detecting a statistically significant effect. Thus, the fact that we continue to observe significant results despite this potential attenuation suggests that our estimates likely understate the true impact of private pressure. To address the related concern that spillovers could induce correlation in the error term, we cluster and bootstrap standard errors at the industry level. The results (untabulated) remain robust.

In untabulated additional analyses, we assess robustness using entropy-balancing (Hainmueller and Xu 2013) and coarsened exact matching (Blackwell, Iacus, King, and Porro 2009). With entropy balancing, the reweighted samples achieve covariate balance between treated and control firms. With coarsened exact matching on two-digit SIC industry and firm size (log market value coarsened into three bins), observations outside common support are discarded, and outcome regressions are estimated on the matched sample. Treated firms remain more likely to remove or explain the sanctioned entity in their CMDs respectively. Overall, the results reinforce our conclusion that private pressure influences firm disclosure behavior.

Additional Intermediate Responses

Following our initial randomized field experiment, we extended the treatment to the original control firms using a randomized rollout (waitlist or stepped-wedge) design, in which all units eventually receive treatment at randomly assigned times (Ditlmann and Paluck 2015). This design offers two advantages: it increases statistical power and external validity by expanding the sample of firms for which disclosure changes can be observed, and it allows us to examine intermediate responses to private inquiries across a broader set of firms. We implemented the control group

treatment between June 26–28, 2024, approximately four weeks after firms filed their 2023 CMDs. Untabulated analyses show that control firms provided longer final responses than treatment firms (150 versus 128 words), while the distribution of final response categories was similar across groups.

VI. CONCLUSION

Section 1502 of the Dodd–Frank Act requires SEC registrants to disclose the sourcing of conflict minerals, with the goal of improving supply-chain transparency. In practice, however, compliance and enforcement have weakened, leaving firms substantial discretion in how they report supply-chain risks. Exploiting U.S. Treasury sanctions on African Gold Refinery (AGR), we study whether private pressure transmitted through firms’ grievance mechanisms influences mandatory disclosure behavior.

Using a randomized field experiment, we show that firms receiving an anonymous, outsider-initiated inquiry are significantly more likely to modify subsequent conflict minerals disclosures. These responses occur along two margins: removal of the sanctioned entity from reported smelter or refiner lists, or retention accompanied by explanatory narrative disclosure. Firms provided with an illustrative disclosure example are more likely to respond through explanation rather than removal, indicating that disclosure models shape how firms adjust reporting choices. We also document substantial engagement with the inquiry itself, with many firms initiating internal reviews or supplier follow-ups.

We further find that disclosure responses vary systematically with proxies for reputational exposure. Firms facing greater public or ESG-related scrutiny are more likely to remove the sanctioned entity, whereas consumer-facing firms are more likely to respond through enhanced narrative explanation. These patterns are consistent with a reputational-cost framework in which

private pressure operates through perceived credibility and anticipated scrutiny rather than public diffusion.

Our study contributes to the literature in three ways. First, we provide causal evidence that private, outsider-initiated pressure transmitted through grievance mechanisms can influence mandatory disclosure choices, extending prior work focused on public pressure or insider whistleblowing. Second, we document that firms respond along multiple disclosure margins and that these responses vary predictably with reputational exposure. Third, we contribute to the growing field-experiment literature in accounting by examining real disclosure outcomes in a regulatory environment characterized by managerial discretion and limited enforcement.

Our findings also inform policy debates surrounding conflict minerals regulation. Consistent with prior evidence of weak enforcement, we document continued disclosure of a sanctioned entity and declining participation in the reporting regime. At the same time, our results suggest that grievance mechanisms can complement formal enforcement by enabling private stakeholders to raise credible concerns that shape disclosure behavior. These insights are relevant to regulators and standard setters, including the SEC, OFAC, GAO, OECD, and the Responsible Minerals Initiative.

We acknowledge several limitations. Our analysis focuses on disclosure responses rather than verified changes in underlying supply-chain relationships, and our setting centers on a single sanctioned entity within the conflict minerals setting. In addition, one channel through which private inquiries may operate is by increasing awareness of sanctioned entities among disclosure gatekeepers. While our design does not distinguish between newly revealed information and increased salience of existing information, the heterogeneous and strategic nature of firms' responses suggests that disclosure revisions reflect deliberation rather than mechanical correction.

Accordingly, our conclusions are confined to how firms adjust disclosures in response to private inquiries and do not speak to the broader effectiveness of sanctions or disclosure mandates. With these caveats, our findings highlight grievance mechanisms as an underexplored but economically meaningful channel through which private stakeholders may influence corporate disclosure behavior.

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FIGURE 1
Timeline of Events and Conflict Mineral Disclosures

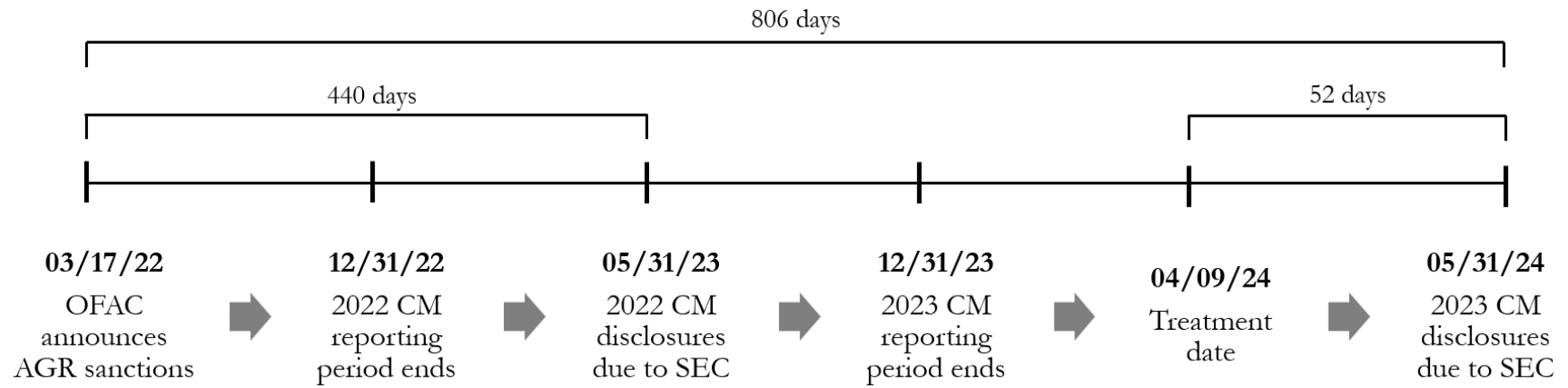


FIGURE 2
Time Trends of Specialized Disclosure Filings with African Gold Refinery (AGR) Partitioned by the Treatment

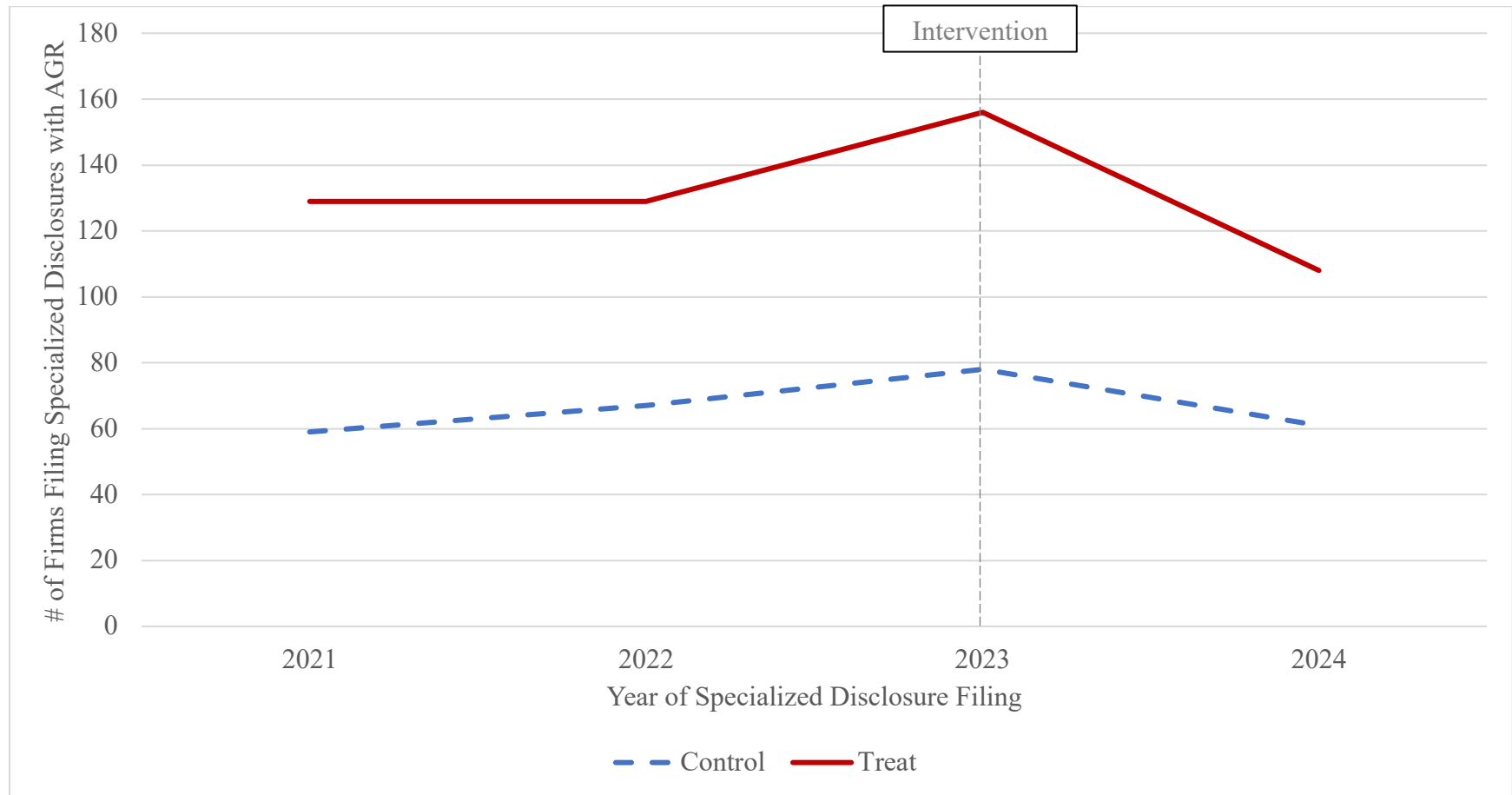


TABLE 1
Sample Selection for Private Pressure Analyses

This table presents the sample selection. Panel A describes how we arrived at our final sample. Panel B provides an industry analysis of the sample. All variables are defined in Appendix A.

Panel A: Sample Selection

Firms with AGR listed in the 2022 CMR Smelter or Refiner Listing (2023 Filing year)	244
Firms with AGR listing that made voluntary disclosures about AGR's sanction	(7)
Firms with AGR Listing utilized in the Pilot Test (5/25(2) and 5/31(1) filings)	(3)
Final Sample	234

Panel B: Industry Analysis

Fama-French 12 Industries	n	%
Consumer Nondurables	1	0.43%
Consumer Durables	24	10.68%
Manufacturing	71	30.34%
Oil, Gas, and Coal Extraction and Production	6	2.56%
Chemicals and Allied Products	5	2.14%
Business Equipment	72	30.77%
Telephone and Television Transmission	3	1.28%
Wholesale, Retail, and Some Services	9	3.85%
Healthcare, Medical Equipment, and Drugs	32	13.25%
Finance	1	0.43%
Other	10	4.27%
Total	234	100%

TABLE 2
Descriptive Statistics for Private Pressure Analyses

This table presents summary statistics for the variables used in our analysis. The sample period of our study is filing year 2023. Panel A provides descriptive statistics of the firm characteristics used in our analyses. Panel B provides descriptive statistics of the firms' grievance mechanisms, hotlines, and signors related to conflict minerals reports. Panel C provides descriptive statistics of the treatment and responses of the sample. Panel D provides descriptive statistics of the treatment and disclosure consequences of the sample. All variables are defined in Appendix A.

Panel A: Firm Characteristics

Variable	N	Mean	SD	P1	Q1	Median	Q3	P99
<i>Revenue (millions)</i>	234	8,809.155	25,584.220	6.026	665.877	2,522.298	6,700.400	156,735.000
<i>Total Assets (millions)</i>	234	13,906.860	41,907.430	40.558	1,012.687	3,285.258	9,444.300	264,037.000
<i>Market Value (millions)</i>	234	22,253.500	128,381.300	23.861	889.216	2,903.498	11,279.640	196,510.200
<i>Sales Growth</i>	234	0.190	0.748	-0.887	0.032	0.102	0.203	2.390
<i>Firm Age</i>	234	29.643	20.381	0.214	12.769	27.125	43.047	96.997
<i>Violation (1/0)</i>	234	0.372	0.484	0.000	0.000	0.000	1.000	1.000
<i>Total Violations</i>	234	1.427	2.803	0.000	0.000	0.000	1.000	12.000
<i>Proportion</i>	234	0.007	0.009	0.003	0.006	0.006	0.006	0.024
<i>Asset 4 Coverage</i>	234	0.885	0.320	0.000	1.000	1.000	1.000	1.000
<i>Glassdoor</i>	234	0.825	0.381	0.000	1.000	1.000	1.000	1.000
<i>Prior Awareness</i>	234	0.010	0.058	-0.087	-0.023	0.000	0.030	0.196
<i>Public Attention</i>	234	2,819.077	19,735.130	0.000	0.000	0.000	838.000	40,719.000
<i>Media Attention</i>	234	2.637	11.612	0.000	0.000	0.000	1.000	80.000
<i>B2C</i>	234	0.047	0.212	0.000	0.000	0.000	0.000	1.000

Panel B: Grievance Mechanisms, Hotlines, and Signors

Variable	N	Mean	SD	P1	Q1	Median	Q3	P99
<i>Grievance Mechanism in SD</i>	234	0.615	0.488	0.000	0.000	1.000	1.000	1.000
<i>Hotline in Code of Conduct</i>	234	0.825	0.381	0.000	1.000	1.000	1.000	1.000
<i>Hotline Website</i>	234	0.812	0.392	0.000	1.000	1.000	1.000	1.000
<i>Hotline Email</i>	234	0.295	0.457	0.000	0.000	0.000	1.000	1.000
<i>Hotline Phone Number</i>	234	0.295	0.457	0.000	0.000	0.000	1.000	1.000
<i>RMI Grievance Mechanism</i>	234	0.047	0.212	0.000	0.000	0.000	0.000	1.000
<i>Legal Signor</i>	234	0.466	0.500	0.000	0.000	0.000	1.000	1.000
<i>CFO Signor</i>	234	0.346	0.477	0.000	0.000	0.000	1.000	1.000
<i>CEO Signor</i>	234	0.043	0.203	0.000	0.000	0.000	0.000	1.000
<i>Other Signor</i>	234	0.145	0.353	0.000	0.000	0.000	0.000	1.000

Panel C: Treatment and Responses

Variable	N	Mean	SD	P1	Q1	Median	Q3	P99
<i>Treat</i>	234	0.667	0.472	0.000	0.000	1.000	1.000	1.000
<i>Treat 1</i>	234	0.333	0.472	0.000	0.000	0.000	1.000	1.000
<i>Treat 2</i>	234	0.333	0.472	0.000	0.000	0.000	1.000	1.000
<i>Days to Final Response</i>	125	15.44	12.198	0.000	3.000	14.000	27.000	43.000
<i>Final Response</i>	156	2.474	1.509	0.000	1.000	3.000	4.000	4.000
<i>Total Word Count</i>	125	106.055	104.041	1.000	34.000	62.000	151.000	464.000

Panel D: Treatment and Disclosure Consequences

Variable	N	Mean	SD	P1	Q1	Median	Q3	P99
<i>Treat</i>	222	0.667	0.472	0.000	0.000	1.000	1.000	1.000
<i>Treat 1</i>	222	0.338	0.474	0.000	0.000	0.000	1.000	1.000
<i>Treat 2</i>	222	0.329	0.471	0.000	0.000	0.000	1.000	1.000
<i>Removed AGR</i>	222	0.234	0.424	0.000	0.000	0.000	0.000	1.000
<i>Explained AGR</i>	222	0.153	0.361	0.000	0.000	0.000	0.000	1.000

TABLE 3
Intermediate Responses

This table summarizes the intermediate responses we received from treated firms before 05/31/2024. The sample period of our study is filing year 2023. Panel A provides the types of responses that we have received while Panel B provides the timeliness and word count of the responses. All variables are defined in Appendix A.

Panel A: Types of Responses

Response	N	%
0 - No Response	31	19.87%
1 - Automated Response	15	9.62%
2 - Human Response with no further actions	8	5.13%
3 - Human Response with investigation	53	33.97%
4 - Human Response with answers	49	31.41%
Total	156	100.00%

Panel B: Timeliness and Word Count of Responses

Response	Days to Respond	Mean Word Count
0 - No Response	-	-
1 - Automated Response	0.333	45.333
2 - Human Response with no further actions	19.125	121.250
3 - Human Response with investigation	15.415	53.442
4 - Human Response with answers	19.490	184.429
Total	15.440	106.055

TABLE 4
Conflict Minerals Disclosures after Treatment

This table examines the potential impact of treatment on firm disclosures in the subsequent reporting period. *Removed_AGR* is equal to one when a firm that included AGR on its smelters and refiners (SoR) list in the 2022 conflict minerals report (CMR) does not include AGR on its 2023 SoR list, and zero otherwise. *Explained_AGR* is equal to one when a firm provides an explanation on AGR or its sanction status in its 2023 CMR, and zero if the firm provides no such explanation. All models are linear probability models with t-statistics reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All other variables are defined in Appendix A.

VARIABLES	(1) <i>Removed_AGR</i>	(2)	(3) <i>Explained_AGR</i>	(4)
<i>Treat</i>	0.088 (1.530)	0.107* (1.845)	0.189*** (4.862)	0.218*** (5.116)
<i>Legal Signor</i>		-0.009 (-0.094)		0.057 (0.791)
<i>CFO Signor</i>		-0.068 (-0.752)		-0.074 (-1.073)
<i>Revenue</i>		-0.016 (-0.475)		0.001 (0.019)
<i>Market Value</i>		0.042 (1.499)		-0.013 (-0.597)
<i>Sales Growth</i>		0.003 (0.090)		0.010 (0.356)
<i>Firm Age</i>		-0.000 (-0.005)		-0.001 (-1.167)
<i>Violations (0/1)</i>		-0.085 (-1.228)		-0.103 (-1.611)
<i>Proportion</i>		-0.786 (-0.543)		-0.719 (-0.581)
<i>Asset 4 Coverage</i>		-0.045 (-0.448)		0.070 (0.947)
<i>Glassdoor</i>		0.164** (2.178)		0.024 (0.332)
<i>Prior Awareness</i>		0.235 (0.515)		0.125 (0.241)
Intercept	0.176*** (3.953)	-0.091 (-0.552)	0.027 (1.427)	0.106 (0.669)
Observations	222	222	222	222
R-squared	0.010	0.058	0.061	0.133
Model	LPM	LPM	LPM	LPM

TABLE 5
Differences in Conflict Minerals Disclosures after Treatment 1 and 2

This table examines the potential impact of the two types of treatment on firm disclosures in the subsequent reporting period. *Removed_AGR* is equal to one when a firm that included AGR on its smelters and refiners (SoR) list in the 2022 conflict minerals report (CMR) does not include AGR on its 2023 SoR list, and zero otherwise. *Explained_AGR* is equal to one when a firm provides an explanation on AGR or its sanction status in its 2023 CMR, and zero if the firm provides no such explanation. *Treat 1* is equal to one if a firm receives the treatment without a disclosure example (Instrument 1 in Appendix C), and zero otherwise. *Treat 2* is equal to one if a firm receives the treatment with a disclosure example (Instrument 2 in Appendix C), and zero otherwise. All models are linear probability models with t-statistics reported in parentheses. The F-test checks the equality of the coefficients of *Treat 1* and *Treat 2*. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All other variables are defined in Appendix A.

VARIABLES	(1) <i>Removed_AGR</i>	(2)	(3) <i>Explained_AGR</i>	(4)
<i>Treat 1</i>	0.104 (1.520)	0.121* (1.722)	0.106** (2.425)	0.138*** (2.797)
<i>Treat 2</i>	0.071 (1.050)	0.093 (1.362)	0.274*** (4.787)	0.298*** (5.184)
<i>Legal Signor</i>		-0.008 (-0.084)		0.052 (0.733)
<i>CFO Signor</i>		-0.067 (-0.746)		-0.078 (-1.144)
<i>Revenue</i>		-0.016 (-0.486)		0.002 (0.089)
<i>Market Value</i>		0.042 (1.516)		-0.015 (-0.687)
<i>Sales Growth</i>		0.003 (0.087)		0.010 (0.391)
<i>Firm Age</i>		-0.000 (-0.046)		-0.001 (-0.814)
<i>Violations (0/1)</i>		-0.084 (-1.201)		-0.111* (-1.802)
<i>Proportion</i>		-0.789 (-0.526)		-0.699 (-0.715)
<i>Asset 4 Coverage</i>		-0.047 (-0.465)		0.080 (1.070)
<i>Glassdoor</i>		0.163** (2.168)		0.029 (0.422)
<i>Prior Awareness</i>		0.248 (0.546)		0.046 (0.093)
Intercept	0.176*** (3.944)	-0.088 (-0.529)	0.027 (1.424)	0.089 (0.553)
Observations	222	222	222	222
R-Squared	0.011	0.059	0.098	0.165
Model	LPM	LPM	LPM	LPM
F-test (p-value)	0.21 (0.647)	0.13 (0.716)	6.29 (0.013)	5.85 (0.016)

TABLE 6
The Role of Public Attention

This table examines the role of public attention on the potential impact of treatment on firm disclosures in the subsequent reporting period. *Removed_AGR* is equal to one when a firm that included AGR on its smelters and refiners (SoR) list in the 2022 conflict minerals report (CMR) does not include AGR on its 2023 SoR list, and zero otherwise. *Explained_AGR* is equal to one when a firm provides an explanation on AGR or its sanction status in its 2023 CMR, and zero if the firm provides no such explanation. *Attention* is equal to one when a firm has above the median 2022 and 2023 of CMR downloads in the pre-period year. All models are linear probability models with t-statistics reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All other variables are defined in Appendix A.

VARIABLES	(1) <i>Removed AGR</i>	(2)	(3) <i>Explained AGR</i>	(4)
<i>Treat</i> × <i>Attention</i>	0.232** (2.004)	0.222* (1.829)	-0.066 (-0.779)	-0.087 (-1.010)
<i>Treat</i>	0.014 (0.199)	0.038 (0.538)	0.210*** (4.499)	0.246*** (4.916)
<i>Attention</i>	-0.129 (-1.550)	-0.084 (-0.742)	0.024 (0.506)	0.192** (2.155)
<i>Legal Signor</i>		-0.009 (-0.094)		0.048 (0.674)
<i>CFO Signor</i>		-0.070 (-0.789)		-0.072 (-1.055)
<i>Revenue</i>		-0.016 (-0.470)		-0.000 (-0.011)
<i>Market Value</i>		0.044 (1.578)		-0.009 (-0.430)
<i>Sales Growth</i>		-0.003 (-0.087)		0.010 (0.355)
<i>Firm Age</i>		-0.001 (-0.409)		-0.004** (-2.000)
<i>Violations (0/1)</i>		-0.089 (-1.278)		-0.095 (-1.505)
<i>Proportion</i>		-0.194 (-0.132)		-0.815 (-0.797)
<i>Asset 4 Coverage</i>		-0.046 (-0.467)		0.080 (1.037)
<i>Glassdoor</i>		0.159** (2.120)		0.036 (0.496)
<i>Prior Awareness</i>		0.245 (0.502)		0.098 (0.203)
Intercept	0.216*** (3.711)	-0.048 (-0.293)	0.020 (1.001)	0.081 (0.530)
Observations	222	222	222	222
R-squared	0.010	0.041	0.061	0.132
Model	LPM	LPM	LPM	LPM

TABLE 7
The Role of Media Attention

This table examines the role of communication channels on the potential impact of treatment on firm disclosures in the subsequent reporting period. *Removed_AGR* is equal to one when a firm that included AGR on its smelters and refiners (SoR) list in the 2022 conflict minerals report (CMR) does not include AGR on its 2023 SoR list, and zero otherwise. *Explained_AGR* is equal to one when a firm provides an explanation on AGR or its sanction status in its 2023 CMR, and zero if the firm provides no such explanation. *Media_Attention* is equal to one if the firm has negative ESG-related media scrutiny in the pre-period year per the RepRisk database greater than the median, and zero otherwise. All models are linear probability models with t-statistics reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All other variables are defined in Appendix A.

VARIABLES	(1) <i>Removed AGR</i>	(2) <i>Removed AGR</i>	(3) <i>Explained AGR</i>	(4) <i>Explained AGR</i>
<i>Treat</i> × <i>Media Attention</i>	0.267** (2.496)	0.238** (2.103)	-0.030 (-0.420)	-0.044 (-0.518)
<i>Treat</i>	0.003 (0.038)	0.033 (0.439)	0.208*** (3.904)	0.234*** (4.152)
<i>Media Attention</i>	-0.166** (-2.168)	-0.164** (-2.045)	-0.036 (-1.428)	0.002 (0.033)
<i>Legal Signor</i>		0.004 (0.038)		0.054 (0.726)
<i>CFO Signor</i>		-0.053 (-0.583)		-0.079 (-1.102)
<i>Revenue</i>		-0.023 (-0.681)		0.003 (0.112)
<i>Market Value</i>		0.047 (1.628)		-0.013 (-0.592)
<i>Sales Growth</i>		-0.003 (-0.084)		0.010 (0.382)
<i>Firm Age</i>		0.000 (0.061)		-0.001 (-1.203)
<i>Violations (0/1)</i>		-0.086 (-1.211)		-0.096 (-1.466)
<i>Proportion</i>		-0.970 (-0.720)		-0.753 (-0.586)
<i>Asset 4 Coverage</i>		-0.032 (-0.323)		0.074 (0.974)
<i>Glassdoor</i>		0.138* (1.813)		0.028 (0.388)
<i>Prior Awareness</i>		0.136 (0.291)		0.153 (0.292)
Intercept	0.218*** (3.882)	-0.030 (-0.175)	0.036 (1.428)	0.083 (0.531)
Observations	222	222	222	222
R-squared	0.029	0.072	0.068	0.135
Model	LPM	LPM	LPM	LPM

TABLE 8
Business to Consumer Industries

This table examines the role of communication channels on the potential impact of treatment on firm disclosures in the subsequent reporting period. *Removed_AGR* is equal to one when a firm that included AGR on its smelters and refiners (SoR) list in the 2022 conflict minerals report (CMR) does not include AGR on its 2023 SoR list, and zero otherwise. *Explained_AGR* is equal to one when a firm provides an explanation on AGR or its sanction status in its 2023 CMR, and zero if the firm provides no such explanation. *B2C* is equal to one if the firm is in a business to consumer industry (i.e., retail and consumer services), and zero otherwise. All models are linear probability models with t-statistics reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. All other variables are defined in Appendix A.

VARIABLES	(1) <i>Removed AGR</i>	(2)	(3) <i>Explained AGR</i>	(4)
<i>Treat</i> × <i>B2C</i>	-0.460 (-1.194)	-0.479 (-1.267)	0.401** (2.078)	0.491** (2.582)
<i>Treat</i>	0.103* (1.767)	0.124** (2.102)	0.171*** (4.365)	0.197*** (4.583)
<i>B2C</i>	0.333 (0.927)	0.307 (0.858)	-0.028 (-1.421)	-0.113* (-1.905)
<i>Legal Signor</i>		-0.008 (-0.083)		0.069 (0.979)
<i>CFO Signor</i>		-0.065 (-0.734)		-0.061 (-0.913)
<i>Revenue</i>		-0.016 (-0.489)		0.001 (0.038)
<i>Market Value</i>		0.046 (1.629)		-0.019 (-0.856)
<i>Sales Growth</i>		0.001 (0.035)		0.018 (0.621)
<i>Firm Age</i>		-0.000 (-0.100)		-0.001 (-0.824)
<i>Violations (0/1)</i>		-0.084 (-1.204)		-0.102 (-1.579)
<i>Proportion</i>		-0.725 (-0.490)		-0.775 (-0.670)
<i>Asset 4 Coverage</i>		-0.055 (-0.542)		0.081 (1.163)
<i>Glassdoor</i>		0.155** (2.070)		0.036 (0.536)
<i>Prior Awareness</i>		0.295 (0.642)		-0.001 (-0.001)
Intercept	0.167*** (3.760)	-0.109 (-0.667)	0.028 (1.421)	0.110 (0.701)
Observations	222	222	222	222
R-squared	0.018	0.068	0.094	0.165
Model	LPM	LPM	LPM	LPM

APPENDIX A

Variable Definitions

Variable	Definition
<i>Revenue (millions)</i>	Total revenues from Compustat
<i>Total Assets (millions)</i>	Total assets from Compustat
<i>Market Value (millions)</i>	Total market value from Compustat
<i>Sales Growth</i>	The change in sales from 2021 to 2022 from Compustat
<i>Firm Age</i>	The number of years prior to December 2022 that the firm has been on CRSP
<i>Violation (1/0)</i>	If the firm has a violation as tracked by violation tracker in the past three years, the indicator is equal to one, and zero otherwise.
<i>Total Violations</i>	A firm's total number of violations from violation tracker for the past three years
<i>Proportion</i>	1 divided by the total number of gold smelters from the 2022 reporting year SD filings.
<i>Asset 4 Coverage</i>	If the firm is covered by Asset 4 regarding suppliers' social impact, the indicator is equal to one, and zero otherwise.
<i>Glassdoor</i>	If the firm is covered by Glassdoor regarding employees' reviews, the indicator is equal to one, and zero otherwise.
<i>Prior Awareness</i>	The average firm's market-adjusted abnormal returns cumulated over a five-day window around the Department of Treasury's announcement of African Gold Refinery's sanctioning.
<i>Public Attention</i>	A firm's 2022 and 2023 conflict minerals report (CMR) downloads from EDGAR in the pre-period year (i.e., 06/01/2022 – 05/31/2023).
<i>Grievance Mechanism in SD</i>	If the firm has a grievance mechanism in its specialized disclosure, the indicator is equal to one, and zero otherwise
<i>Hotline in Code of Conduct</i>	If the firm has a hotline in its code of conduct, the indicator is equal to one, and zero otherwise
<i>Hotline Website</i>	If the firm has a hotline website, the indicator is equal to one, and zero otherwise
<i>Hotline Email</i>	If the firm has a hotline email, the indicator is equal to one, and zero otherwise
<i>Hotline Phone Number</i>	If the firm has a hotline phone number, the indicator is equal to one, and zero otherwise
<i>RMI Grievance Mechanism</i>	If the firm references the Responsible Minerals Initiative's grievance mechanism, the indicator is equal to one, and zero otherwise
<i>Legal Signor</i>	If the specialized disclosure indicates a legal signor as the person responsible for the conflict minerals report, the indicator is one, and zero otherwise
<i>CFO Signor</i>	If the specialized disclosure indicates a CFO signor as the person responsible for the conflict minerals report, the indicator is one, and zero otherwise

<i>CEO Signor</i>	If the specialized disclosure indicates a CEO signor as the person responsible for the conflict minerals report, the indicator is one, and zero otherwise
<i>Other Signor</i>	If the specialized disclosure indicates a signor other than Legal, CFO, or CEO as the person responsible for the conflict minerals report, the indicator is one, and zero otherwise
<i>Treat</i>	If a firm receives an instrument, the indicator is one, and zero otherwise
<i>Treat 1</i>	A binary variable equal to one if a firm receives treatment without a disclosure example (Instrument 1 in Table 2 Panels C and D and Appendix C), and zero otherwise
<i>Treat 2</i>	A binary variable equal to one if a firm receives treatment with a disclosure example (Instrument 2 in Table 2 Panel C and D and Appendix C), and zero otherwise
<i>Days to Final Response</i>	The number of days it takes to provide a final response to the inquiry as of the filing deadline of 5/31/2024
<i>Final Response</i>	A variable ranging from 0 to 4 in which 0 represents no response, 1 represents an automated response, 2 represents a human response with no further actions, 3 represents a human response with investigation, and 4 represents a human response with answers
<i>Total Word Count</i>	The number of words in the final response
<i>Sentiment</i>	The sentiment of the words in the final response
<i>Explained_AGR</i>	A binary variable equal to one if a firm's 2023 CMR includes AGR on its list of smelters or refiners and there is an explanation for, or reference to, the AGR sanctions in the firm's report (in main text or footnotes), and zero if there is no explanation.
<i>Removed_AGR</i>	A binary variable equal to one if a firm's 2023 CMR removes AGR either from its list of smelters or refiners (SoR) or elsewhere in the report, and zero if AGR is mentioned in the report
<i>Attention</i>	A binary variable equal to one if a firm's 2022 and 2023 CMR downloads from EDGAR in the pre-period year (i.e., 06/01/2022 – 05/31/2023) are above the median, and zero otherwise.
<i>Media_Attention</i>	A binary variable equal to one if a firm's media coverage per the RepRisk database in the pre-period year (i.e., 06/01/2022 – 05/31/2023) are above the median, and zero otherwise. Reprisk reflects exposure to negative ESG-related media coverage.
<i>B2C</i>	A binary variable equal to one if the firm is in a business to consumer industry (i.e., retail [SIC 5200 - 5999] and consumer services [SIC 7000 – 7999]), and zero otherwise.

APPENDIX B
Firms' SDs with Explanations on African Gold Refinery

#	Name	CIK	Filing Date	Explanation
1	TESLA, INC.	1318605	2023-05-31	** Certain Tier 1 suppliers reported the presence of this entity that was sanctioned by the United States Department of Treasury, Office of Foreign Assets Control on March 17, 2022, specifically, CID003185 - African Gold Refinery. Because of the over-reporting nature of the industry CMRT information collection process, and the nature of the supply chains and goods, we are unable to confirm this, or any, SoR is or was active in our supply chain.
2	FORD MOTOR CO	37996	2023-05-31	*** Entity became an OFAC Specially Designated National (SDN) on March 17, 2022. Ford does not have reason to believe African Gold Refinery Limited ("AGR", CID03185) was present in Ford's supply chain after the SDN designation.
3	AMPHENOL CORP /DE/	820313	2023-05-31	African Gold Refinery is an eligible gold refiner listed in the RMI facility database but has been removed from the RMI RMAP due to sanction related activities which were announced by the U.S. Department of the Treasury's Office of Foreign Assets Control on March 17, 2022. Pursuant to this action, we initiated an assessment to determine the presence of African Gold Refinery across our organization and are actively implementing measures to transition from this entity. In addition, as a result of the London Bullion Market Association ("LBMA") removing six Russian-based gold refiners from its Good Delivery List on March 7, 2022, we carried out measures to assess our supply chain exposure and transition from these entities in 2022. Amphenol did not source directly from either African Gold Refinery or the Russian-based gold refiners in 2022 as Amphenol is a downstream user of 3TGs. Universal Precious Metals Refining Zambia and Phoenix Metal Ltd. are both not eligible smelters or refiners according to the RMI RMAP and we are continuing to actively transition from these entities.
4	ORMAT TECHNOLOGIES, INC.	1296445	2023-05-30	+ On March 17, 2022, the U.S. Department of the Treasury, Office of Foreign Assets Control (OFAC) sanctioned this gold refinery in Uganda. Ormat has no direct business with the refinery. Through our due diligence examination, we identified our direct suppliers who reported utilizing the Ugandan refinery in their supply chain. We are in communication with the direct suppliers who listed this refinery to confirm if the refinery actually exists in our supply chain and will continue as necessary to follow-up to have this refinery removed from our supply chain.

#	Name	CIK	Filing Date	Explanation
5	CANON INC	16988	2023-05-30	* On March 17, 2022, the U.S. Department of the Treasury added this gold refinery in Uganda to the OFAC's Specially Designated Nationals (SDN) List. The Canon Group has no direct business with the refinery. We alerted the first-tier suppliers of the Canon Group not to use this gold refinery, and requested that suppliers utilizing this refinery in their supply chains provide us with schedules for eliminating the refinery therefrom ("elimination schedules"). As of the date of this report, we remain in the process of confirming the elimination schedules from our suppliers.
6	APTIV PLC	1521332	2023-05-26	* The United States Department of Treasury, Office of Foreign Assets Control ("OFAC") designated African Gold Refinery ("AGR") as a Specially Designated National ("SDN") on March 17, 2022, which falls within the reporting period of this CMR. Pursuant to its standard policy, Aptiv complies with all applicable sanctions laws. To date, Aptiv has not received confirmation from any of its suppliers that they utilized AGR in providing 3TG to Aptiv. Based on a review conducted by Aptiv upon AGR's designation as an SDN, Aptiv did not identify any direct transactions or contractual relationships with AGR. Aptiv continues to work with its supply base to ensure compliance with all applicable sanctions laws.
7	ALLOT LTD	1365767	2023-05-22	*** On March 17, 2022, the U.S. Department of the Treasury, Office of Foreign Assets Control (OFAC) sanctioned this gold refinery in Uganda. Allot has no direct business with the refinery. Through our due diligence examination, we identified our first-tier manufacturers and suppliers who reported utilizing the Ugandan refinery in their supply chain. We are in communication with the first-tier manufacturers and suppliers who listed this refinery to confirm if the refinery actually exists in our supply chain and will continue as necessary to follow-up to have this refinery removed from our supply chain.

APPENDIX C

Treatment Instruments and Submission Channels

Instrument 1

Dear XXX,

I have identified that XXX has African Gold Refinery (AGR), a sanctioned entity, in its supply chain.

On March 17, 2022, the U.S. Department of the Treasury's Office of Foreign Assets Control (OFAC) sanctioned the AGR in Uganda because it had been involved in the illicit movement of gold valued at hundreds of millions of dollars per year from the Democratic Republic of the Congo (DRC). This illicit movement of gold provides revenue to armed groups that threaten the peace, security, and stability of the DRC. This action was taken pursuant to Executive Order 13413, which targets, among other things, individuals and entities involved in activities that threaten the peace, security, or stability of the DRC or that undermines democratic processes or institutions in the DRC. Please see <https://home.treasury.gov/news/press-releases/jy0664> for additional details about the sanction. AGR, the sanctioned entity, can be found in your most recent conflict minerals report for 2022. I have the following questions:

1. Do you have any exemptions (e.g., from OFAC or Department of Justice) to continue doing business with AGR?
2. Does your company have any disclosure updates that I am unaware of?
3. If not, can you tell me what the company is doing to address the sanctioned entity in their supply chains?

Thank you for your time and attention to this matter.

Instrument 2 (Information Incremental to Instrument 1)

... Please see <https://home.treasury.gov/news/press-releases/jy0664> for additional details about the sanction. AGR, the sanctioned entity, can be found in your most recent conflict minerals report for 2022, which has no detailed explanation about this sanction.

Tesla, for example, included a description of the sanction in their CMR report as “**** Certain Tier 1 suppliers reported the presence of this entity [African Gold Refinery] that was sanctioned by the United States Department of Treasury, Office of Assets Control on March 17, 2022, specifically, CID003185 - African Gold Refinery. Because of the over-reporting nature of the industry CMRT information collection process, and the nature of the supply chains and goods, we are unable to confirm this, or any, SoR is or was active in our supply chain. Tesla is in communication with Tier 1 suppliers who listed this SoR, and will continue necessary follow-up to have this SoR removed from their supply chain. Overall, we continue to engage with our Tier 1 suppliers to improve due diligence efforts and transparency.*”.

I have the following questions...

Submission Channels

This section provides implementation details on how treatment alerts were submitted to firms and documents the reporting channels used. Treatment alerts were delivered through firms' designated grievance mechanisms, consistent with how firms solicit compliance and ethics concerns from internal and external stakeholders. For 129 of the 156 treated firms (82.7 percent), alerts were submitted through web-based ethics hotlines provided in firms' conflict minerals disclosures or codes of conduct. Of these submissions, 123 alerts (78.8 percent of the treated sample) were transmitted via third-party incident-management platforms, and 6 alerts (3.8 percent) were submitted through proprietary firm-managed web portals. For the remaining 27 firms (17.3 percent), which did not maintain functional web-based reporting systems, alerts were submitted via firm-designated email addresses associated with ethics, compliance, whistleblowing, or related reporting functions. These addresses were obtained from firms' conflict minerals reports or codes of conduct.

Third-party platforms used include EthicsPoint (Navex), Convercent, Mitrastech Syntrio, Whistleblowerservices.com, Integrity Counts, Safe Hotline, and SAI360. These providers state that reports are anonymized and transmitted securely without revealing identifying information about the reporter. Consistent with these policies, our submissions did not require personal identifiers, and we did not disclose our identity or institutional affiliation in any communication.

Firm responses are consistent with the alerts being perceived as anonymous. In several cases, firm representatives explicitly acknowledged the reporter's anonymity and indicated that internal review or investigation would proceed on that basis. When follow-up communication was requested, firms offered mechanisms that preserved anonymity, such as message exchanges within the reporting platform or anonymized phone communication. No firm indicated that it could identify the sender of the alert. Table C1 summarizes the distribution of treatment alerts across reporting channels.

Table C1: Treatment Submission Channels

Submission Channel	Number of firms	Percent
Third-party web-based ethics hotlines	123	78.8%
Firm-managed web portals	6	3.8%
Email-based reporting addresses	27	17.3%
Total	156	100%

APPENDIX D

Examples of Intermediate Responses

Automatic Response

Thank you for contacting the EthicsLine service. We have received your report and this matter is now pending.

Human Response with No Further Action

The investigation of this matter has been completed, and appropriate action has been taken. Due to the confidential nature of any investigation, findings must remain confidential and cannot be shared. Thank you for filing your report.

Human Response with Investigation

CommScope appreciates the concerns you have raised regarding your workplace. Based on the information you have provided, we have commenced an investigation, which is ongoing at this time. We will post a further update on the CommAlert system as soon as the investigation is concluded. In the meantime, if you have additional information that you believe we should consider or would like to discuss this matter further, please use the Follow-up section on the CommAlert AlertLine. Thank you.

Human Response with Specific Answers

Hello,

Thank you for your message. Sanmina takes compliance with applicable law seriously and investigates all reports of potential violations of law. As your questions below, we provide the following answers:

1. Do you have any exemptions (e.g., from OFAC or Department of Justice) to continue doing business with AGR?

Sanmina does not do business with AGR. As part of Sanmina's conflict minerals due diligence process, suppliers upstream of Sanmina have reported the existence of AGR in their supply chains.

2. Does your company have any disclosure updates that I am unaware of?

We will be filing our 2023 conflict minerals report with the SEC by May 31, 2024.

3. If not, can you tell me what the company is doing to address the sanctioned entity in their supply chains?

To the extent our conflict minerals due diligence process shows this smelter supplier (a) remains in Sanmina's indirect supply chain and (b) remains sanctioned, we will bring this information to the attention of the supplier(s) that identified this smelter in their reporting to us and request that they remove this smelter from their supply chain.

APPENDIX E
Examples of Effects of Private Pressure in Firms' SDs with Explanations on African Gold Refinery

#	Name	CIK	Filing Date	Explanation
1	FLowservE Corp	30625	2024-05-31	*Certain of our suppliers reported the presence of the following SORs in their supply chains: African Gold Refinery, which was sanctioned by the United States Department of Treasury, Office of Foreign Assets Control (“OFAC”) on March 17, 2022, JSC Uralelectromed, which was sanctioned by OFAC on July 20, 2023, and OJSC “The Gulidov Krasnoyarsk Non-Ferrous Metals Plant” (OJSC Krastsvetmet), which was sanctioned in the United Kingdom on November 8, 2023. Because of the over-reporting nature of the industry CMRT information collection process, and the nature of our supply chains, we are unable to confirm whether these, or any, specific SORs are or were active in our supply chains. Our policy is to comply with all applicable sanctions laws, and we do not have any direct transactions or contractual relationships, nor are we, to the best of our knowledge, engaging in indirect transactions, with these entities.
2	EASTMAN KODAK CO	31235	2024-05-31	* Certain Tier 1 suppliers reported the presence of this entity that was sanctioned by the United States Department of Treasury, Office of Foreign Assets Control on March 17, 2022, specifically Smelter ID CID003185 – African Gold Refinery. Due to the over-reporting nature of the industry CMRT information collection process, and the nature of the supply chains and goods, we are unable to confirm this, or any, smelters or refiners is or was active in our supply chain.
3	MOOG INC	67887	2024-05-31	Moog is aware that African Gold Refinery is subject to U.S. Department of the Treasury’s Office of Foreign Assets Control sanctions. Moog does not purchase directly from, or otherwise interact directly with, this entity. Moreover, many suppliers provide a company-level CMRT, meaning that the information Moog receives for conflict minerals due diligence is overinclusive and cannot provide conclusive evidence that African Gold Refinery is actually within Moog’s supply chain.
4	EMCORE CORP	808326	2024-05-31	Certain suppliers reported the presence of this entity that was sanctioned by the United States Department of Treasury, Office of Foreign Assets Control on March 17, 2022, specifically, CID003185 - African Gold Refinery (“AGR”). Because of the over-reporting nature of the industry CMRT information collection process, and the nature of the supply chains and goods, we are unable to confirm AGR was or is active in our supply chain.

#	Name	CIK	Filing Date	Explanation
5	WOODWARD, INC.	108312	2024-05-31	Certain suppliers reported potential sourcing from a gold smelter in the Uganda that has not been assessed by RMI (CID003185 – African Gold Refinery)(“AGR”). On March 17, 2022, the U.S. Department of the Treasury added African Gold Refinery in Uganda to the OFAC’s Specially Designated Nationals (SDN) List. Because of the over-reporting nature of the industry CMRT information collection process, and the nature of the supply chains and goods, we are unable to confirm if minerals from AGR are in our products. Woodward is in contact with suppliers who reported potential sourcing from AGR and will continue to engage with those suppliers to improve due diligence efforts and transparency.
6	CALIX, INC	1406666	2024-05-31	Certain surveyed suppliers reported potential sourcing of 3TG in 2023 from CID003185 - African Gold Refinery (“AGR”). To our knowledge, AGR is a gold smelter in a Covered Country and is not currently listed as a conformant or active facility by the RMI. Furthermore, on March 17, 2022, the United States Department of Treasury’s Office of Foreign Assets Control designated AGR as a Specially Designated National. Calix has not identified any direct transactions or contractual relationships with AGR, and to date has not received confirmation from any of our suppliers that they utilized 3TG sourced from AGR in any of Calix’s products in 2023. Calix continues to work with our suppliers to ensure improved transparency and accountability in our supply chain.